Beyond Code: An SRE Agent for Self-Healing Operations

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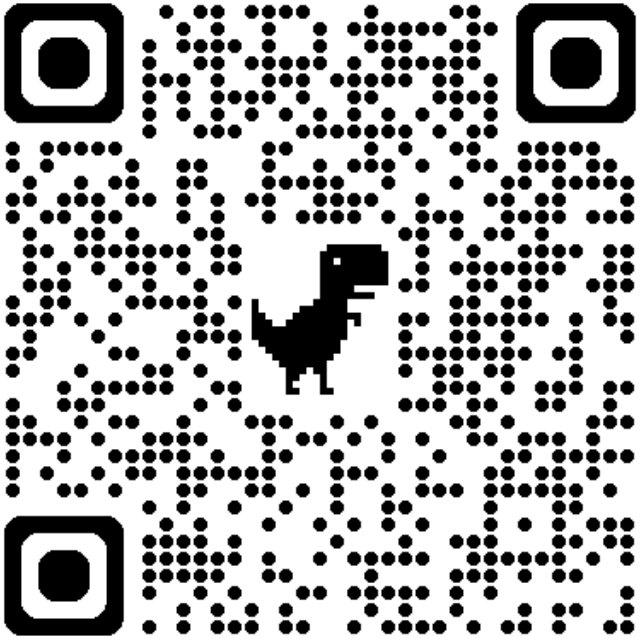
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https://www.4tecture.ch/events/dwx26-sre-agent



Agenda

- Demo first: incident, analysis, fix
- Why build our own SRE agent?
- Architecture for Agentic Operations
- Demo code, step by step
- Production guardrails

Cold Open: Pods Turn Red

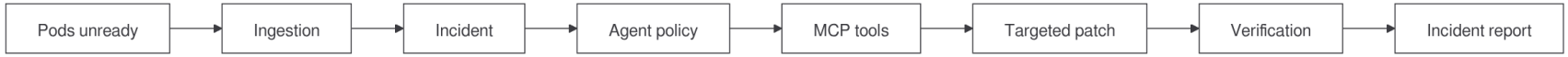
- Inject a FastFood deployment error
- Kubernetes reports unhealthy pods
- SRE agent starts automatically
- Agent analyzes, patches, verifies

Demo



Pods go red

What Just Happened



Why This Matters

- Operations often becomes toil
- Diagnosis creates context switching
- Runbooks quietly go stale
- Good fixes require evidence
- Repetition calls for automation

AGENTIC OPERATIONS

From Chatbot To Operator

- Chat: answers questions
- Assistant: suggests next steps
- Agent: uses tools autonomously
- Operator: acts within policy
- SRE agent: evidence-first, recovery-focused

Autonomy Is A Control

Lower

- ReadOnly
- SuggestFix
- PostmortemOnly

Higher

- ApprovalRequired
- AutoRemediate
- Policy-gated writes

The Evidence Contract

- Read live state
- Ground hypotheses in sources
- Plan the smallest safe change
- Document before and after
- Verify recovery actively

Why Not Just Buy It?

- There are SRE agents services available for Azure and other clouds
- Customized agent gives more control over signals, tools, and policies
- We are buildint it to learn what is actually required for a safe, evidence-first SRE agent

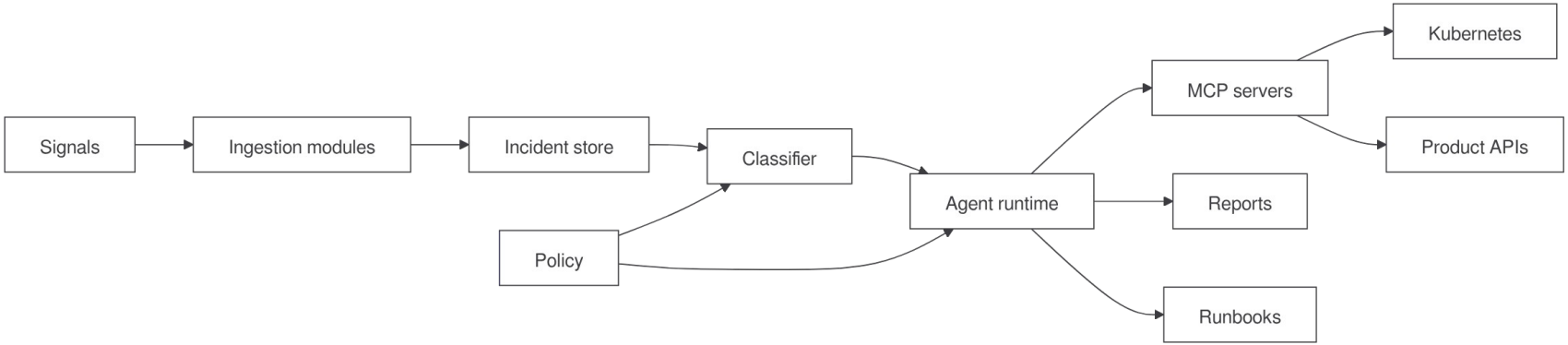
Azure SRE Agent

- Queries Application Insights and identifies a memory trend that started 40 minutes before the alert
- Correlates the trend with a deployment event from your GitHub repository two hours earlier
- Identifies the specific commit and proposes two mitigations: restart the affected pod and adjust the memory scaling threshold (HPA)
- Creates a ticket in ServiceNow, PagerDuty, or your incident channel with the full investigation summary prefilled

Source: https://learn.microsoft.com/azure/sre-agent/overview

ARCHITECTURE

Reference Architecture



Ingestion: Where Incidents Start

- Kubernetes pod health
- Prometheus or Grafana alerts
- Log and trace anomalies
- ChatOps or manual escalation
- Scheduled health checks

Incident Core

- Receive signal
- Deduplicate fingerprint
- Apply rule-based triage
- Build mode-specific policy
- Write timeline and events

Tools: MCP As The Operations Interface

- Standardized tool descriptions
- Servers encapsulate target systems
- Agent discovers available tools
- Policies constrain tool usage
- Audit per tool call

Skills Are Runbooks For Agents

- Version product knowledge
- Block dangerous shortcuts
- Encode known failure modes
- Require explicit verification
- Review them like code

DEMO CODE

Repository Map

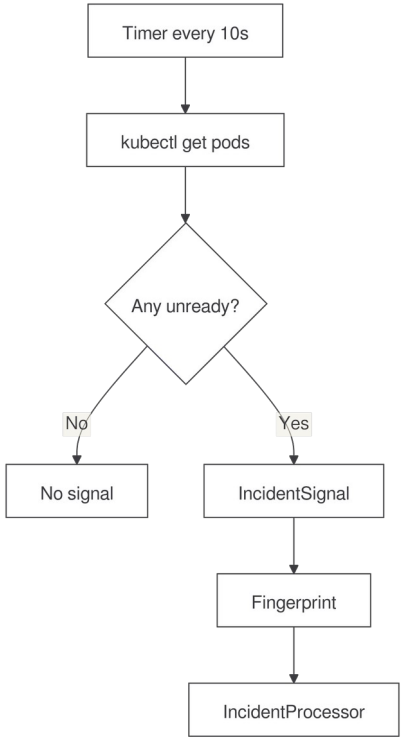
- `SreAgent.Core`: incidents, policies, ingestion
- `SreAgent.Agent.Copilot`: runtime adapter
- `SreAgent.Web`: dashboard and prompts
- `mcp-setup/k8s`: ServiceAccount and RBAC
- `skills/`: operational runbooks

Demo



How it was built

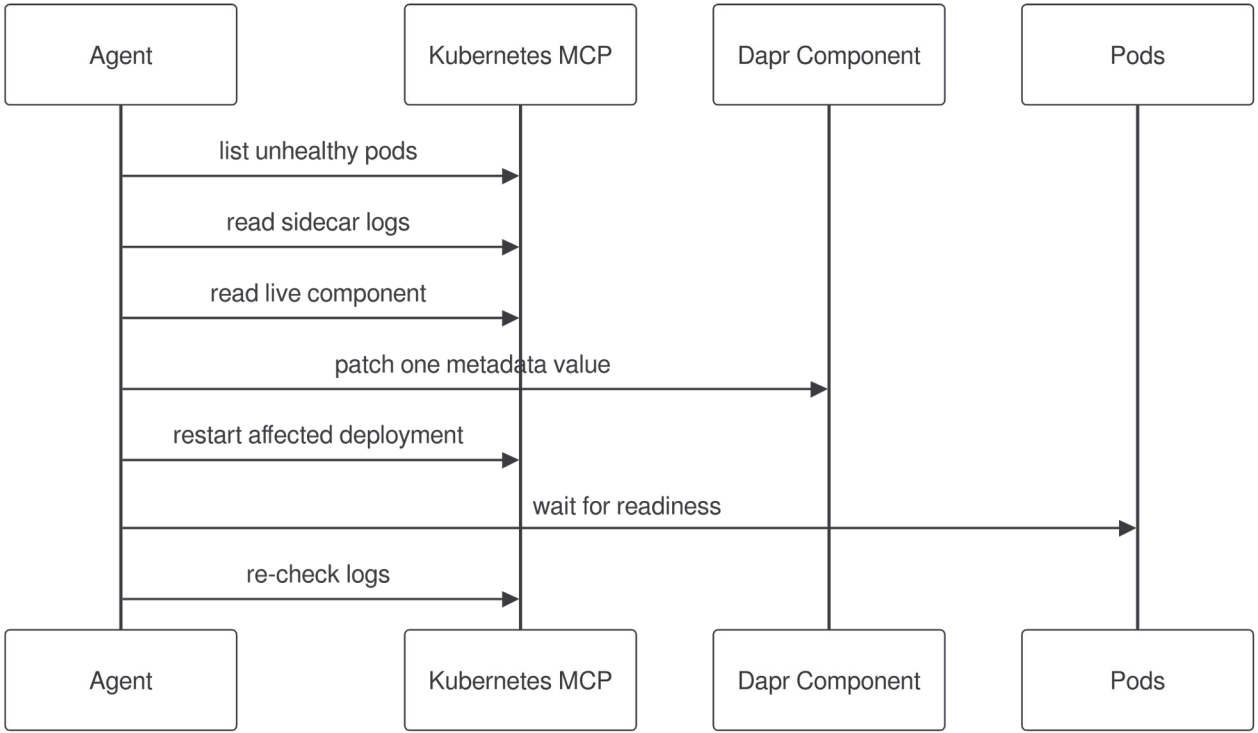
Kubernetes Ingestion



Prompt: The Operating Contract

- Restore service safely
- Prefer tools over guesses
- Read before changing
- Patch minimally
- Verify readiness and logs
- Produce incident report

Remediation Sequence in our Demo



Incident Report As Artifact

- Signals
- Diagnosis
- Actions Taken
- Verification
- Post-incident

GUARDRAILS

Least Privilege In Kubernetes

- Dedicated ServiceAccount
- RoleBindings per namespace
- Only required resources
- Only required verbs
- No cluster-admin shortcut

Tool Approval Is Policy

- Allowlist MCP servers
- Denylist risky tools
- Require approval before writes
- Sandbox host access
- Log every tool call

Prompt Injection Is Operational Reality

- Logs contain untrusted input
- Tickets can contain instructions
- Alert payloads are data, not commands
- Tools need hard boundaries
- Reports need sources and evidence

Self-Healing Requires Design

- Redundancy reduces risk
- Bulkheads limit blast radius
- Automatic actions must be idempotent
- Rollback must be known first
- Recovery must be tested

Decision Rule For Auto-Fix

- Is the symptom unambiguous?
- Is root cause evidenced?
- Is the fix small and reversible?
- Are permissions minimal?
- Is verification automatable?

RECAP

Takeaways

- SRE agents need more than a model
- Evidence beats confidence
- MCP makes tools portable
- Skills make runbooks actionable
- Autonomy should grow gradually
- Guardrails are product features

Thank you! Let’s continue the conversation

If you would like to discuss your challenges or explore how we can support you, please contact us.

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ProAuth

ReaFx

References

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- Microsoft Learn - Agent Framework: https://learn.microsoft.com/agent-framework/overview/
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- Azure Well-Architected - Self-healing: https://learn.microsoft.com/azure/well-architected/reliability/self-preservation
- GitHub Docs - Copilot CLI: